

Introduction

As we know, the search engines use booleans. Using Boolean in the retrieval of information is a very old technique. In the earlier period, it was used in this case as exact matching words in the query to match documents. Therefore, it is the classical method and the most widely used model. It uses the classic set of theories and Boolean logic. In this way, it will search to see if the query words meet the boolean requirement in the query.

Here, indexing makes it easy for the search engine. The word indexing is used to express or describe the document. It involves the Boolean operators as

1. **AND:** Using the add operator gives a result from a document having the specified keyword. For example, if we give mango and banana, it will search both key words to filter the relevant information.
2. **OR:** The OR operator makes wide searches such that at least one is specified in the keywords. For example, in rat or bat, it will search either for the word rat, bat, or both as it expands the range of searching.
3. **NOT:** The NOT operator excludes the specific keywords that are used in the search. For example, if we use mango, not fruit, it will give the result that is related to mango and exclude other fruits so that it makes use of searching for the specific part only.

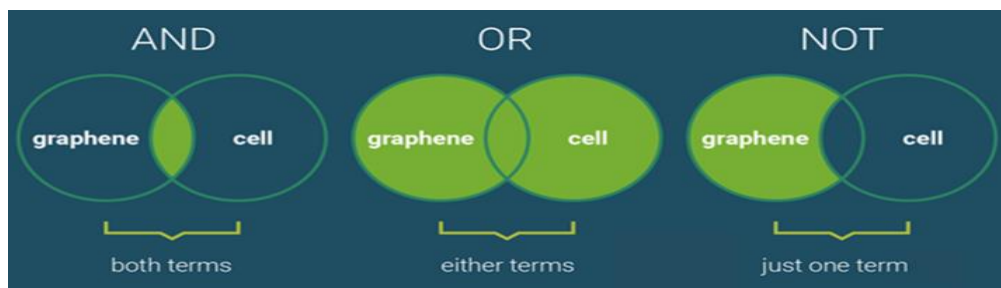


Fig: Power of Boolean Search

Boolean query processing:

1. It finds the location of the term.
2. For each term, it gets the entire row.
3. Processing will do an operation on each term pair.

terms							
he	drink	ink	likes	pink	thing	wink	
2	1	0	2	0	0	1	← D1
1	3	0	1	0	0	0	← D2
1	1	1	1	0	1	0	← D3
1	1	1	1	1	0	0	← D4
1	1	1	1	1	0	1	← D5

Fig: Dictionary Postings

It has faced many critiques, as I explained herein below:

1. While working with precision and recall, the main critique is the accuracy verses recall trade-off. Using boolean techniques, the accuracy is high because it searches for the key word to match in the document. But recall may be low from this technique as it may have missed or ignored the relevant document.
2. Complexity: The user may have to face the complexity issues in order to have a better understanding of the boolean logic. It may be difficult for some user to determine the order of the operation while searching. Therefore, complex queries are misunderstood.
3. Limited semantic understanding: Due to its limited understanding, it will skip the relevant document in cases where the query is not exactly matched.
4. Scalability: When we work on a large document, it has less chance for scalability and is inefficient to work on. In a large document, the process will be very lengthy and take more time for computation while working with the large index.